

ARK-442 — RESULTS

Authorization Boundary Degradation Under Verification-to-Execution Delay on IBM Quantum (Heron r2)

Remnant Fieldworks Inc. — Derek Hone

Governing principle: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

VERDICT: PASS

All four preregistered PASS conditions (Field 20) were met with the in-situ SPAM baseline within ceiling on both qubits and no calibration drift. Expired and replayed authorizations did **not** execute the payload; fresh reverification restored ALLOW.

Execution record

Item	Value
Backend	ibm_marrakesh (Heron r2, 156 qubits)
Instance	open-instance
Selected qubit pair (Field 10)	Q_A = 5 (RE 0.4883%), Q_P = 6 (RE 0.4028%), sum 0.8911%
Qualifying connected pairs	92 (min-sum rule; next best [140,141]=0.9033%)
Shots	8,192 / arm × 8 arms (principal); 2,048 × 4 (SPAM)
Transpiler	optimization_level=1, initial_layout=[5,6], no dynamical decoupling
SPAM job id	d9clmusjeosc73fgeo10
Principal job id	d9clom4jeosc73fgeq3g
Calibration snapshot	calibra- tion_snapshot_marrakesh_20260716.json

Ordering honored (Field 27): code+MANIFEST committed (lock 74212534) → qubit pair frozen (86e0432e) → SPAM committed (4aef0834) → principal job id committed **before** reading (1a58d554) → raw counts committed (63a5dceb) → analysis (this record).

Preregistered metrics (Field 17) with 95% Wilson CIs

Metric	Definition	Value	95% CI
S_A_0	ALLOW, 0 μ s (Arm 1)	0.9878	[0.9852, 0.9900]
S_A_short	ALLOW, $\sim$ 0.5 μ s (Arm 2)	0.9884	[0.9858, 0.9905]
S_A_medium	ALLOW, $\sim$ 1.0 μ s (Arm 3)	0.9866	[0.9838, 0.9888]
S_A_long	ALLOW, $\sim$ 2.0 μ s (Arm 4)	0.9832	[0.9801, 0.9857]
L_expired	expired-auth leakage (Arm 5, PRIMARY)	0.0001	[0.0000, 0.0007]
L_replayed	replayed-auth leakage (Arm 6, PRIMARY)	0.0015	[0.0008, 0.0026]
S_reverified	reverified ALLOW fidelity (Arm 7)	0.9916	[0.9894, 0.9933]
SPAM_baseline	idle readout (Arm 8)	0.0001	[0.0000, 0.0007]
Δ_B	S_A_0 – L_expired	0.9877	—
L_expired_corrected	L_expired – SPAM_baseline	0.0000	—
L_replayed_corrected	L_replayed – SPAM_baseline	0.0013	—

In-situ SPAM job (separate): Q_A p01 = 0.0039, Q_P p01 = 0.0005 (both $\leq$ 0.02). SPAM drift vs Arm 8 idle = 0.0004 ($\leq$ 0.005). Gate **passed**.

Decision against preregistered rule (Fields 20-22)

PASS condition	Requirement	Observed	Met?
Expired DENY	$L_{\text{expired_corrected}} \leq 0.02$	0.0000	✓
Replayed DENY	$L_{\text{replayed_corrected}} \leq 0.02$	0.0013	✓
Reverification	$S_{\text{reverified}} \geq 0.90$	0.9916	✓
Boundary discrimination	$\Delta_B \geq 0.70$	0.9877	✓
In-situ SPAM	≤ 0.02 on both qubits	0.0039 / 0.0005	✓
Calibration drift	≤ 0.005	0.0004	✓

→ **PASS.**

Delay-decay characterization (H2a — descriptive, non-gating)

ALLOW survival vs verification-to-execution delay:

Delay	S_A	95% CI
0.0 μs	0.9878	[0.9852, 0.9900]
0.5 μs	0.9884	[0.9858, 0.9905]
1.0 μs	0.9866	[0.9838, 0.9888]
2.0 μs	0.9832	[0.9801, 0.9857]

The ALLOW survival erodes by ≈ 0.5 percentage points from 0 $\rightarrow$ 2.0 μs — a small, decoherence-consistent decline (the payload T1 on Q_P is long relative to the 2 μs probe window, so the boundary degrades only slightly on this timescale). The strict point-estimate ordering is not monotone (Arm 2 sits marginally above Arm 1, within overlapping CIs — statistical noise at the 8,192-shot level), so the preregistered monotonicity flag (H2a) reads **False**; H2a is descriptive and carries no pass/fail weight (Fields 6, 20). See [plots/delay_decay.png](#).

Secondary hypotheses (Bonferroni, Field 19)

- **H2b (replay $\approx$ expiry):** L_{replayed} (0.0015) and L_{expired} (0.0001) — 99% Wilson intervals **overlap** $\rightarrow$ a replayed stale bit is statistically no better than an expired one. Consistent with H2b.

- **H2c (reverification concordance):** $S_{\text{reverified}}$ (0.9916) and S_{A_0} (0.9878) — 95% Wilson intervals **overlap** → fresh reverification is concordant with the immediate ALLOW reference. Consistent with H2c.

Interpretation boundaries (Field 23)

This is a **metrological characterization of decoherence-driven erosion** of a verify-then-execute authorization boundary as the verification-to-execution separation increases. It is **NOT new physics** and **NOT a cryptographic claim**. “Leakage” means unauthorized payload activation, not computational-basis leakage. Findings apply only to this boundary implementation on this qubit pair ($Q_A=5$, $Q_P=6$) on `ibm_marrakesh` at this calibration (2026-07-16) and do not generalize without replication. The delay arms measure how ALLOW survival decays with idle time; the expired/replayed arms show a stale authorization is correctly not honored; the reverification arm shows ALLOW is restored with a fresh measurement.

Relationship to prior work (Field 24)

ARK-442 extends the single $\sim 1 \mu\text{s}$ stale-auth arm of ARK-441 into a delay-resolved characterization (0/0.5/1.0/2.0 μs) and adds explicit expired-auth, replay-after-expiry, and reverification arms. The DENY-arm thresholds ($L_D_{\text{corrected}} \leq 0.02$, $\Delta_B \geq 0.70$) carry over from ARK-441 and are met here. ARK-446 (cross-device replication on `ibm_marrakesh`, same $Q_A=5/Q_P=6$ pair) returned PASS concurrently; ARK-442 characterizes the boundary’s degradation under delay on that same pair.

ARK-442 Results — Remnant Fieldworks Inc. / Derek Hone — 2026-07-16. Verdict PASS. Metrological characterization of decoherence-driven boundary erosion; not new physics, not a cryptographic claim. No Rescue After Failure.